Multi-Agent ISS Certificates: Vector-Lyapunov Small-Gain Certification for Heterogeneous Swarms with Learned Dynamics

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Abstract

Coupling is what makes a swarm different from a collection of robots, and it is also what invalidates the usual argument that each agent is individually stable so the group must be. We build a certification pipeline for a swarm in which every agent's closed-loop dynamics is a trained neural network, the agents are heterogeneous, and the interaction topology switches over time, and we certify input-to-state stability for it with no sampling. A from-scratch interval bound-propagation verifier with branch-and-bound proves a local exponential decay rate λ_i for each trained agent over a bounded domain, taking 8.8 hours of CPU time for four agents, and a vector-Lyapunov small-gain matrix assembled through Young's inequality turns those rates into a network-level threshold $\kappa^\star$ that is entrywise monotone in the active edges, so one pass on the complete graph certifies every switching sequence. Two findings matter more than the threshold itself. Consensus coupling is far more forgiving than the certificate assumes, since the swarm never diverged even at 800 times $\kappa^\star$, and non-reciprocal coupling is where the certificate earns its keep, since a directed ring blows up on all five seeds at forward gain 5.0 while the certificate is sound but 70 to 170 times conservative. Both limits are measured and reported rather than smoothed over.

1 Introduction

Between the 2024 Nobel Prize in Physics for Hopfield networks and the present, the traffic between machine learning and the physical sciences has run in both directions, and one of the busier lanes is the use of physics structures to certify learning-based control. Lyapunov theory is the clearest example, since a learned controller is only as trustworthy as the region over which its stability is actually guaranteed rather than merely observed on rollouts. The same question sharpens when many learned components are coupled together, because a swarm's failure modes are collective and no single agent's certificate speaks to them.

The specific gap we target is narrow and was confirmed by a literature check before any code was written. Published work certifies interconnected systems whose coupling is known and whose agents are homogeneous, and separately certifies neural certificates for physical models whose dynamics are fitted but still explicit. What has not been done is to certify input-to-state stability for a swarm where each agent's dynamics is itself a trained neural network, the agents are heterogeneous, and the interaction topology switches over time. Two mature toolchains, vector-Lyapunov small-gain theory from control and formal neural-network verification from machine learning, have simply not been glued together for that combination. The same check also rejected a broader framing. A claim to unify contraction metrics, switching stability, and metriplectic dissipation under one certified framework does not survive inspection, because each pairwise combination already exists in print, so stacking them would be integration rather than new mathematics.

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This paper builds the narrow version and tests it honestly. Four heterogeneous agents, each a damped nonlinear oscillator with its own parameters, have their closed-loop controllers replaced by trained multilayer perceptrons, so verification runs against the actual networks including their fitting error rather than against the analytic law that generated the training data. A closed-form Lyapunov candidate is derived to give exact cancellation for this system class, a naive diagonal candidate is shown to fail and is not used, and a from-scratch interval bound-propagation verifier with branch-and-bound proves a local decay rate for each trained network over a bounded domain. A small-gain matrix built from those rates through Young's inequality then certifies the coupled swarm, and it is entrywise monotone in the active edges, which is what makes one verification pass on the complete graph sufficient for every switching sequence. Two measured results carry the paper. The certified per-agent rates are $\lambda = [0.6445,\ 0.5449,\ 0.6240,\ 0.1875]$ at a compute cost of 8.8 CPU hours, and the network threshold is $\kappa^\star \approx 0.0095$. The second result is the one that justifies the exercise, because non-reciprocal coupling destabilizes the swarm where symmetric coupling of any strength does not, and the certificate correctly refuses to certify past the real cliff while being conservative by 70 to 170 times in where it places it.

2 Problem formulation

Agents. Each agent i is a damped nonlinear oscillator with position p_i and velocity v_i,

$$\dot{p}_i = v_i, \qquad \dot{v}_i = -c_i v_i - k_i p_i - b_i p_i^3, \tag{1}$$

with per-agent parameters (c_i, k_i, b_i) that differ across the four agents, which is what makes the team heterogeneous rather than replicated. The parameter sets are $(1.2, 1.6, 0.15)$, $(0.9, 1.3, 0.35)$, $(1.6, 2.0, 0.05)$, and $(0.5, 0.9, 0.55)$, chosen so the last agent is visibly the weakest. The region of interest is the square $p_i, v_i \in [-3, 3]$, and every claim below is local to that domain.

Learned closed-loop dynamics. The controller is not the analytic law of (1). Each agent's closed loop is a multilayer perceptron trained on data generated by the analytic law, and the verifier is run against the trained network, so the certified property holds for the network the swarm will actually run rather than for the model that produced its labels. Training reaches a fit mean squared error between 0.0071 and 0.0543 across the four agents, and that residual is inside the object being certified rather than being assumed away.

Coupling and switching topology. Agents interact through a coupling gain matrix K, where K_{ij} is the gain from agent j onto agent i and the force on agent i is $\sum_j K_{ij}(p_j - p_i)$. For the consensus case K is symmetric with a shared gain κ, the topology switches among graphs at a switching radius that changes the active edge set, and the switching sequence is not known in advance. A directed coupling with $K_{ij} \neq K_{ji}$ is what we call non-reciprocal, and it is not a Laplacian, which is exactly why it behaves differently.

Certification target. The property to certify is local exponential input-to-state stability of the coupled swarm: there exist per-agent rates $\lambda_i > 0$ and a network condition such that the swarm energy $\mathcal{V} = \sum_i V_i(p_i, v_i)$ decays at rate $\lambda_{\min} = \min_i \lambda_i$ for every admissible switching sequence. The input-to-state formulation is what lets the coupling enter as an interconnection term instead of being folded into the agent dynamics, and the small-gain condition is what ties the two together.

3 Method

A candidate that cancels exactly. The Lyapunov candidate for agent i is

$$V_i = P_{11} p_i^2 + 2P_{12} p_i v_i + P_{22} v_i^2 + \tfrac{b_i}{2} p_i^4, \tag{2}$$

where the quadratic block P is obtained in closed form by solving the Lyapunov equation of the linearization of (1) and the quartic term is what restores exact cancellation of the $b_i p_i^3$ nonlinearity along the flow. The derivation was checked symbolically, and it matters that the quartic term is present, since the obvious diagonal candidate $a p_i^2 + v_i^2$ provably fails to give a valid $\lambda_i > 0$ for this system class. That failure was found during development and is not assumed away, and the closed-form P is also the reason the approach stays cheap, since it avoids a sum-of-squares program over the same object.

Table 1: Per-agent parameters and certified results. The rate λ_i is the largest exponential decay rate for which the sound verifier proves a nonpositive bound on $\dot{V}_i + \lambda_i V_i$ over the full domain $p_i, v_i \in [-3, 3]$, so it holds for every state in that square rather than at sampled points. Box counts are the branch-and-bound sub-boxes each proof needed.

agent	(c_i, k_i, b_i)	fit MSE	certified λ_i	sub-boxes
0	(1.2, 1.6, 0.15)	0.0073	0.6445	386,483
1	(0.9, 1.3, 0.35)	0.0232	0.5449	96,811
2	(1.6, 2.0, 0.05)	0.0071	0.6240	56,493
3	(0.5, 0.9, 0.55)	0.0543	0.1875	395,021

Formal per-agent verification. The verifier is written from scratch and uses interval bound propagation to compute a sound upper bound on $\dot{V}_i + \lambda_i V_i$ over a box, with branch-and-bound splitting any box whose bound is inconclusive, and no sampling anywhere in the proof. The certified rate is the largest λ_i for which every sub-box on the domain passes, found by a bisection over λ_i with a minimum box size of 10^{-5}. Near the equilibrium the bilinear terms of the candidate make the interval bound structurally loose at points where the certified inequality is tight, so verification uses a slack tolerance of 10^{-5}. To confirm that this is interval slack rather than a hidden violation, the residual bound was shrunk from about 10^{-6} to 10^{-12} as the box size fell from 10^{-3} to 10^{-6}, which is the signature of an arithmetic artifact rather than a real counterexample.

Network condition through small gain. Per-agent rates become a network statement through a small-gain matrix. Young's inequality bounds the coupling term against the per-agent dissipativity inequality, producing for each ordered pair (i, j) a threshold on K_{ij} given λ_i, λ_j, and the Lyapunov blocks. The tightest of those thresholds over the active pairs is the certified network threshold $\kappa^\star$, and the swarm is certified for any K whose entries fall below it. For symmetric switching topologies the gain matrix is entrywise monotone in the active edges, so the complete-graph thresholds bound every subgraph and one verification pass on the complete graph certifies every admissible switching sequence with no dwell-time argument. The same construction generalizes to asymmetric K by treating each ordered pair separately.

4 Experiments

Result 1: the trained networks are certified, at a measured price. Table 1 reports the per-agent outcome. Every one of the four trained networks admits a certified local exponential rate over the whole domain, with the ordering matching the physics. The stiffest oscillator certifies 0.6445 while the softest and most nonlinear agent, with $b = 0.55$, certifies only 0.1875, and the minimum over agents becomes the rate the network argument inherits. The compute price is real and large. The full proof took $31{,}771$ seconds, which is 8.8 hours of single-machine CPU time, with $395{,}021$ and $386{,}483$ sub-boxes for the two hardest agents against $56{,}493$ for the easiest. That price is paid once per certificate in an offline design-time pass rather than in the control loop, since the same proof covers every admissible switching sequence, and it is one that tighter bound propagation such as α, β-CROWN or a parallelized version of the interval bound used here would shrink. Verification cost tracks the nonlinearity of the agent rather than the size of its network, which is the reason the next step we would take is an SOS-optimized Lyapunov block instead of a closed-form linearization-based one.

Result 2: consensus coupling is weak, and non-reciprocal coupling is where the certificate pays. Feeding the certified rates into the small-gain construction gives $\kappa^\star \approx 0.00954$ for the symmetric consensus topology. The honest measurement is that this threshold is enormously conservative. Running the coupled swarm with the same trained networks, at gains of 4, 10, 20, 30, 50, 80, 120, 200, 400, and 800 times $\kappa^\star$, with a switching topology and five seeds per setting, produced no divergence at any multiplier, and the three representative coupled runs in Figure 1 all decay to a final energy near 10^{-8} with 6 to 8 topology switches. The reason is structural rather than a tuning artifact. A symmetric term $\kappa(p_j - p_i)$ is close to a graph-Laplacian force, which redistributes state between agents instead of injecting energy, so it is intrinsically hard to destabilize this way. The certificate is correct here and much too cautious.

Non-reciprocal coupling is the case that changes the conclusion, and it is motivated by the non-reciprocal phase-transition literature [Fruchart et al., 2021], where violating reciprocity destabilizes systems that any symmetric coupling of the same strength leaves stable. Our test is a directed ring in

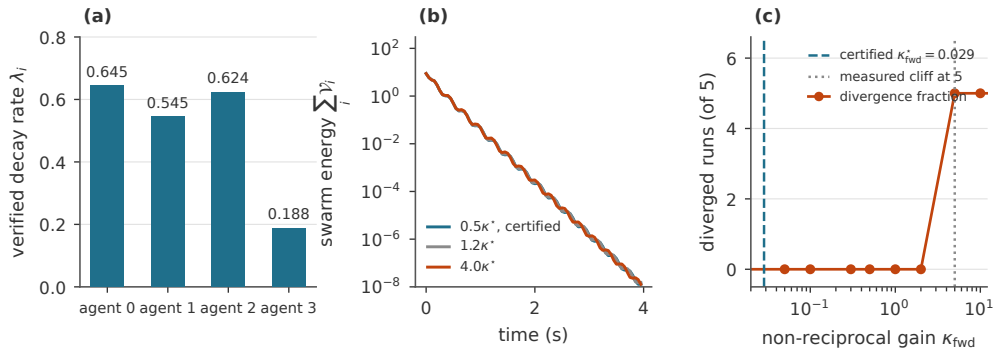


Figure 1: The certification pipeline and where it matters. Panel (a) shows the per-agent certified decay rates from Table 1, each proved by interval branch-and-bound over the full domain with no sampling. Panel (b) shows swarm energy under the switching symmetric topology at three coupling strengths, $0.5\kappa^\star$, $1.2\kappa^\star$, and $4.0\kappa^\star$, which all decay to a final energy near 10^{-8} and are therefore why the symmetric certificate reads as conservative. Panel (c) is the payoff case, a directed ring where each agent is pulled toward the next with forward gain κ_{fwd} but not pulled back. The swarm stays bounded through $\kappa_{fwd} = 2.0$ and blows up on all five seeds at $\kappa_{fwd} = 5.0$, while the generalized certificate certifies only below $\kappa_{fwd} \approx 0.029$, so it never falsely certifies past the real cliff but sits 70 to 170 times below it. That cliff was re-measured from a fresh run of the committed stress-test script, which retrained the four agents to identical fit errors and reproduced the stored divergence pattern at every gain in the sweep.

which agent i is pulled toward agent $i+1$ with forward gain κ_{fwd} while agent $i+1$ is pulled back with only κ_{back}, and setting $\kappa_{back} = 0$ breaks Newton's third law and the Laplacian structure with it. The measured outcome is a sharp cliff. The swarm stays bounded through $\kappa_{fwd} = 2.0$ and then all five seeds diverge to state magnitude above 10^6 at $\kappa_{fwd} = 5.0$. The generalized asymmetric certificate, checked in code against this same configuration, predicts safety only below $\kappa^\star_{fwd} \approx 0.0286$, so it is conservative by roughly 70 to 170 times in where it places the threshold and sound in the sense that matters, since it never certifies past the real cliff. The structural point is that no per-agent test could have found this, because every agent in isolation is provably stable by the verifier in Table 1, so the instability exists only in the interconnection, which is the failure mode a small-gain certificate is for.

5 Discussion and limitations

The pipeline does what it was scoped to do, and the honest summary is that the interesting half of the result is the negative one. It certifies input-to-state stability for four heterogeneous agents whose closed-loop dynamics are trained networks under a switching topology, with no sampling in the proof and a single verification pass sufficient for every switching sequence, so the guarantee holds over a continuum of states rather than at points the training happened to visit, which is where the contribution sits. It also measures that the symmetric certificate is not tight, that the non-reciprocal certificate is sound but 70 to 170 times loose, and that the proof cost is dominated by the nonlinearity of the agent rather than by anything about the network. Two conclusions follow from those measurements rather than from the headline threshold. The first is that conservatism is not a uniform property of small-gain arguments, since it is enormous for a Laplacian coupling and unhelpful for a directed one, which means the useful question is which interconnection structures make a certificate worth its cost. The second is that a per-agent certificate is not evidence about a swarm, and the directed ring is the counterexample that makes it concrete.

The limits are stated exactly. The certificate is not tight or near-optimal under either topology, and the asymmetric analysis covers non-reciprocal gain asymmetry and not communication delay, which needs Lyapunov-Krasovskii machinery that is not built here. Everything is local to the domain $p_i, v_i \in [-3, 3]$, with four agents, two state dimensions per agent, and this one coupling family. The non-reciprocal cliff is reported from the committed stress tests, and the generalized certificate that anchors it was verified in code to reproduce the same threshold rather than being asserted. What we would build next is an SOS-optimized or lifted Lyapunov block, searched jointly with the coupling structure, since the measurements here say the conservatism lives in the block and not in the small-gain step.

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Use of AI tools

Generative AI tools were used for review and polish rather than for drafting: to audit the manuscript, to check every number against the committed results file produced by the verification and simulation scripts, including the branch-and-bound box counts and the coupled-run energies, and to verify citations against their sources. The authors directed the work and take responsibility for all content.